

① Find the flow of $\dot{x} = -2(x-5)$. Global attractor?

Let $\eta \in \mathbb{R}$ and consider the IVP $\dot{x} = -2(x-5)$, $x(0) = \eta$

$\dot{x} = -2(x-5) \Leftrightarrow \dot{x} + 2x = 10 \Rightarrow$ first-order linear ^{non}homog. d.e. with c.c.

$\rightarrow$ general sol: x_h .

$\dot{x} + 2x = 0$ has the general solution $x_h = c \cdot e^{-2t}$, $c \in \mathbb{R}$.

$x + 2 = 0 \Rightarrow x = -2 \Rightarrow e^{-2t} \Rightarrow$

$\rightarrow$ particular sol: x_p

$\dot{x} + 2x = 10$ has a constant particular solution: $x_p = 5$

$5' + 2 \cdot 5 = 10 \checkmark$

$\Rightarrow$ general solution: $x = x_h + x_p = c \cdot e^{-2t} + 5$, $c \in \mathbb{R}$. $\left. \begin{array}{l} x(0) = \eta \Rightarrow c + 5 = \eta \Rightarrow c = \eta - 5 \end{array} \right\} \Rightarrow$

$\Rightarrow \varphi(t, \eta) = (\eta - 5) \cdot e^{-2t} + 5$, $(\forall) (t, \eta) \in \mathbb{R}^2$. (THE FLOW)

$\lim_{t \rightarrow \infty} \varphi(t, \eta) = \lim_{t \rightarrow \infty} \left(\frac{\eta - 5}{e^{2t}} + 5 \right) = 0 + 5 = 5$, $(\forall) t \in \mathbb{R}$. $\left. \begin{array}{l} \varphi(t, 5) = 5, (\forall) t \in \mathbb{R} \end{array} \right\} \Rightarrow$

$\Rightarrow$ the equilibrium point $\eta^* = 5$ is a global attractor. (because if the limit equals the same equilibrium for every η , that equilibrium is a global attractor)

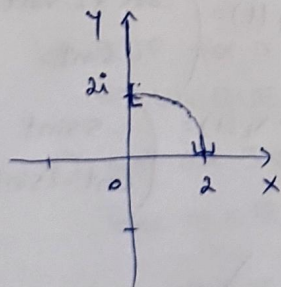
Equilibrium points: $\dot{x} = -2(x-5) = 0 \Rightarrow x = 5$.

② Represent in the complex plane the curves:

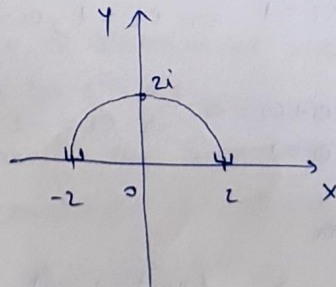
$\{2 \cdot e^{it} : t \in [0, \frac{\pi}{2}]\}$, $\{2 \cdot e^{it} : t \in [0, \pi]\}$, $\{2 \cdot e^{it} : t \in [0, 2\pi]\}$

$\downarrow$
 $\eta: \{ \eta \in \mathbb{R} \mid \lim_{t \rightarrow \infty} \varphi(t, \eta) = \eta^* \}$
 $\hookrightarrow$ global att.

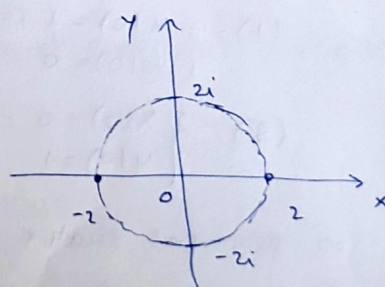
$z = 2 \cdot e^{it} = 2(\cos t + i \sin t) \Rightarrow |z| = 2$
 $\arg(z) = t \quad \left. \begin{array}{l} \end{array} \right\} \Rightarrow$ each curve is a part of the circle centered in the origin of radius 2.



$t \in [0, \frac{\pi}{2}]$



$t \in [0, \pi]$



$t \in [0, 2\pi]$

$$\textcircled{3} A = \begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix}$$

a) Principal matrix solution of $x' = AX$

→ General solution of the system $(=)$ $\begin{cases} x' = 2x - 5y \\ y' = x - 2y \end{cases}$, $X = \begin{pmatrix} x \\ y \end{pmatrix}$
REDUCTION METHOD.

$$\begin{cases} x' = 2x - 5y \\ y' = x - 2y \end{cases} \Rightarrow x = y' + 2y$$

$$y'' = x' - 2y' = (2x - 5y) - 2y' = 2(y' + 2y) - 5y - 2y' = -y$$

$\Rightarrow y'' = -y \Rightarrow$ second-order lin. homog. d.e. w c.c.

$$\Rightarrow y'' + y = 0$$

$$k^2 + 1 = 0 \Rightarrow k^2 = -1 \Rightarrow k_{1,2} = \pm i \rightarrow \cos t, \sin t$$

$$\Rightarrow y(t) = c_1 \cos t + c_2 \sin t$$

$$y'(t) = -c_1 \sin t + c_2 \cos t$$

$$\begin{aligned} x(t) &= y'(t) + 2y(t) = -c_1 \sin t + c_2 \cos t + 2 \cdot (c_1 \cos t + c_2 \sin t) \\ &= \cancel{3c_2 \cos t} - c_1 \cdot (2 \cos t - \sin t) + c_2 \cdot (\cos t + 2 \sin t) \end{aligned}$$

$\Rightarrow$ General sol:

$$\begin{cases} x = c_1 \cdot (2 \cos t - \sin t) + c_2 \cdot (\cos t + 2 \sin t) \\ y = c_1 \cos t + c_2 \sin t \end{cases} \quad c_1, c_2 \in \mathbb{R}$$

→ 2 IVP's for the principal matrix sol:

$$x' = AX, \quad x(0) = 1, \quad y(0) = 0 \quad (1)$$

and

$$x' = AX, \quad x(0) = 0, \quad y(0) = 1$$

$$x(0) = c_1 \cdot 2 + c_2$$

$$y(0) = c_1$$

$$(1): \begin{cases} x(0) = 1 \\ y(0) = 0 \end{cases} \Leftrightarrow \begin{cases} 2c_1 + c_2 = 1 \\ c_1 = 0 \end{cases} \Rightarrow c_2 = 1, c_1 = 0 \Rightarrow x_1(t) = \begin{pmatrix} \cos t + 2 \sin t \\ \sin t \end{pmatrix}$$

$$(2): \begin{cases} x(0) = 0 \\ y(0) = 1 \end{cases} \Leftrightarrow \begin{cases} 2c_1 + c_2 = 0 \\ c_1 = 1 \end{cases} \Rightarrow c_2 = -2, c_1 = 1 \Rightarrow x_2(t) = \begin{pmatrix} -5 \sin t \\ \cos t - 2 \sin t \end{pmatrix}$$

$\Rightarrow$ principal matrix solution:

$$E(t) = \begin{pmatrix} x_1(t) & x_2(t) \\ \vdots & \vdots \end{pmatrix} = \begin{pmatrix} \cos t + 2 \sin t & -5 \sin t \\ \sin t & \cos t - 2 \sin t \end{pmatrix}$$

a) $a, b \in \mathbb{R}^?$ s.t. $H: \mathbb{R}^2 \rightarrow \mathbb{R}$, $H(x, y) = x^2 + ay^2 + bxy$ global first integral of $x' = AX$

b) e^{At} ?

e^{At} = THE PRINCIPAL (unique) matrix solution of $x' = AX$

$$\Rightarrow e^{At} = E(t).$$

b) c) $a, b \in \mathbb{R}^?$ s.t. $H: \mathbb{R}^2 \rightarrow \mathbb{R}$, $H(x, y) = x^2 + ay^2 + bxy$ global f.i. of $x' = AX$.

$$\begin{cases} x' = 2x - 5y \\ y' = x - 2y \end{cases}, A = \begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix}$$

$H \in C^1(\mathbb{R}^2)$, $(\forall) a, b \in \mathbb{R}$.

$$H \text{ global f.i. } (\Leftrightarrow) \frac{\partial H}{\partial x}(x, y) \cdot (2x - 5y) + \frac{\partial H}{\partial y}(x, y) \cdot (x - 2y) = 0, (\forall) (x, y) \in \mathbb{R}^2.$$

$$\frac{\partial H}{\partial x}(x, y) = 2x + by$$

$$\frac{\partial H}{\partial y}(x, y) = 2ay + bx$$

$$\Rightarrow (2x + by)(2x - 5y) + (2ay + bx)(x - 2y) = 0, (\forall) (x, y) \in \mathbb{R}^2.$$

$$\Rightarrow 4x^2 - 10xy + 2bxy - 5by^2 + 2axy - 4ay^2 + bx^2 - 2bxy = 0$$

$$\Rightarrow (4+b)x^2 + (2a-10)xy - (5b+4a)y^2 = 0.$$

$$\Rightarrow \begin{cases} 4+b=0 \\ a-5=0 \\ 5b+4a=0 \end{cases} \Rightarrow \begin{cases} b=-4 \\ a=5 \end{cases}$$

$$\Rightarrow H(x, y) = x^2 + 5y^2 - 4xy \Rightarrow \text{global f.i.}$$

④ How many solutions has the following pb.?

a) $x''' + t^2 x = 0$, $x(0) = x'(0) = 1 \Rightarrow$ third order, 2 conditions $\Rightarrow$ NOT an IVP

b) $x''' + t^2 x = 0$, $x(0) = x'(0) = x''(0) = 1 \Rightarrow$ third order, 3 cond $\Rightarrow$ IVP.

b) IVP for a linear d.e. of third order with non-ct. coeff. $a(t) = t^2 \in C(\mathbb{R})$
 $\Rightarrow$ UNIQUE SOLUTION in $C^3(\mathbb{R})$.

a) not an IVP. $\Rightarrow$ make it an IVP:

Take $\eta \in \mathbb{R}$, add condition $x''(0) = \eta \Rightarrow$ IVP

$\Rightarrow$ unique solution $\varphi(\cdot, \eta)$

$\Rightarrow$ as many solutions as points in $\mathbb{R}$. (all η).

5). $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{50} x(100-x)$ (DISCRETE DYN. SYSTEM)

a) fixed points? stability.

DON'T DIVIDE BY X.

$$\boxed{f(x) = x} \Leftrightarrow \frac{1}{50} x(100-x) = x \Rightarrow \frac{1}{50} (100-x) = 1 \Rightarrow 100-x = 50 \Rightarrow x = 50.$$

$$\Leftrightarrow 100x - x^2 = 50x \Leftrightarrow x^2 - 50x = 0 \Leftrightarrow x(x-50) = 0.$$

$$\Rightarrow \eta_1^* = 0, \eta_2^* = 50$$

$$f(x) = \frac{1}{50} x(100-x) = \frac{100x}{50} - \frac{x^2}{50} = 2x - \frac{x^2}{50}$$

$$\Rightarrow f'(x) = 2 - \frac{2x}{50}$$

$$|f'(0)| = 2 > 1 \Rightarrow \eta_1^* = 0 \text{ unstable}$$

$$|f'(50)| = 0 < 1 \Rightarrow \eta_2^* = 50 \text{ attractor.}$$

b) Basin of attraction of the attractor fixed point.

$\rightarrow$ graph of f .

x	$-\infty$	0	50	100	∞
f'	+	+	+	0	- - - - -
f	$-\infty$	$\nearrow 0$	$\nearrow 50$	$\searrow 0$	$\searrow -\infty$

$\eta_1^* = 0$ is a fixed point ($f(0) = 0$)

$\eta_2^* = 50$ is a fixed point ($f(50) = 50$)

$\eta = 100$ not a fixed point: $x_1 = f(100) = 0$
 $x_2 = f(0) = 0$

Basin of attraction of 50 (attractor):

$$A_{50} = \{ \eta : \lim_{n \rightarrow \infty} x_n = 50 \} = (0, 100).$$

$$\left. \begin{array}{l} x_0 = 100 \\ x_1 = f(x_0) = f(100) = 0 \\ x_2 = f(x_1) = f(0) = 0 \\ x_3 = f(x_2) = f(0) = 0 \\ \vdots \end{array} \right\} \Rightarrow \lim_{n \rightarrow \infty} x_n = 0 \neq 50$$